

Ashish Misra, Ph.D.

Assistant Professor

Department of Biotechnology

Indian Institute of Technology, Hyderabad

Kandi, Sangareddy-502285, Telangana

E-mail: ashishmisra@bt.iith.ac.in

Telephone: +91-9494688000(cell)

ACADEMIC APPOINTMENTS

Assistant Professor, Department of Biotechnology

October 2016 onwards

Indian Institute of Technology Hyderabad

EDUCATION

- | | |
|---|------|
| Ph.D., Molecular Biophysics Unit,
Indian Institute of Science, Bangalore, India
Thesis titled "Unusual Acylation Properties of Type II Fatty Acid Biosynthesis Acyl
Carrier Proteins". | 2008 |
| M.Sc. Biochemistry, First Class
University of Lucknow, India | 2002 |
| B.Sc. in Chemistry, Botany, Zoology, First Class
University of Lucknow, India | 2000 |

RESEARCH EXPERIENCE

- | | |
|---|-----------------------------|
| - University of Massachusetts Medical School, USA | December 2008- October 2016 |
| - Indian Institute of Science, India. | August 2003 – November 2008 |
| - Jawaharlal Nehru University, New Delhi. | August 2002 – March 2003 |

PUBLICATIONS

- Srivastava D, Gupta A, Verma N, **Misra A**. Widespread Prevalence of CD19 Exon 5-6 Skipping in Primary Pediatric B- Cell Acute Lymphoblastic Leukemia Patients. *Molecular and Cellular Pediatrics* (accepted).
- Srivastava D, **Misra A**. Long Non-Coding RNAs in B-cell acute lymphoblastic leukemia: Disease implication, challenges and therapeutic opportunities. *Non-coding RNA Research*, 2026, 16, 40-56.
- Gupta A, Verma N, **Misra A**. Novel case of Coexisting TCF3 and IGH Gene Rearrangement in Pediatric B-cell Acute Lymphoblastic Leukemia– favourable response to steroids and standard chemotherapy. *Indian Journal of Pathology and Microbiology*, 2025, Volume 68, 0377-4929.
- Srivastava D, Nayak V, Pahari S, G Sandeep, **Misra A**. Epitranscriptomic regulation of immunity: The role of m6A in shaping immune response dynamics. *Cytokine*, 2025, 194, 157011.
- Gupta P, Srivastava D, Nayak V, Rao V, Susheela K, Sharma R, Rajappa S, Mishra P, **Misra A**. YTHDC1 orchestrates oncogenic splicing via the CLK1-SRSF1 splicing machinery to regulate castration-resistant prostate cancer progression. *Molecular and Cellular Biochemistry*, 2025, 480, 5157–5172
- Raut R, Srivastava D, Saini T, Chakraborty A, Bais M, Mishra P, **Misra A**. Integrative Analysis Identifies FBXO5 as a Critical Mediator of CRPC Progression and Bone Metastatic Potential. *Discover Oncology*, 2025, 16, 1495.

- Raut R, Srivastava D, Nayak V, Saini T, Gupta P, Chakraborty A, Choudhury C, Bais M, Mishra P, **Misra A**. Clinical Data Investigation Identifies MARK3 as an Oncogenic Driver in Castration-Resistant Prostate Cancer. *Biochemistry and Biophysics Reports*, 2025, 42, 102003.
- Saini T, Srivastava D, Raut R, Mishra P, **Misra A**. IGF2BP2 promotes castration resistant prostate cancer progression by regulating the AR-V7 mRNA stability. *Cancer Reports*, 2025, 8, e70096.
- Muley A, Kumbhakar S, Raut R, Mathur S, Roy I, Saini T, A, **Misra A*** & Maji S*. Mononuclear Copper (II) complexes with Polypyridyl Ligands: Synthesis, Characterization, DNA Interactions/Cleavages and in vitro Cytotoxicity towards Human Cancer Cells. *Dalton Transactions*, 2024, 53, 11697-11712 (* **corresponding authors**)
- Saini T, Gupta P, Raut R, Nayak V, Bharathnaveen P, Mishra P, **Misra A**. AR-V7 Expression Facilitates Accelerated G2/M Phase Transition in Castration-Resistant Prostate Cancer. *Experimental Cell Research*, 2024, 438(1) 114026.
- Srivastava D, **Misra A**. Computational Analysis to Elucidate Parkinson's Disease Pathophysiology and Therapeutics. 2024 *IEEE International Conference on Contemporary Computing and Communications (InC4)* DOI: 10.1109/InC460750.2024.10649206
- Kumbhakara S, Gupta P, Giri B, Karumbana KS, Muleya A, **Misra A***, Maji S*. Photolability of NO in ruthenium nitrosyls with pentadentate ligand induces exceptional cytotoxicity towards VCaP, 22Rv1 and A549 cancer cells under therapeutic condition. *Journal of Molecular Structure*, 2022, 1265, 133419 (* **corresponding authors**)
- Karumbana KS, Raut R, Gupta P, Muleya A, Giri B, Kumbhakara S, **Misra A***, Maji S*. Mononuclear cobalt(II) complexes with Polypyridyl Ligands: Synthesis, Characterization, DNA interactions and in vitro cytotoxicity towards human cancer cells. *Journal of Inorganic Biochemistry*. 2022, 233, 111866 (* **corresponding authors**)
- Karumbana KS, Muleya A, Raut R, Gupta P, Giri B, Kumbhakara S, **Misra A***, Maji S*. Mononuclear Co(II) polypyridyl complexes: synthesis, molecular structure, DNA binding/cleavage, radical scavenging, docking studies and anticancer activities. *Dalton Transactions*, 2022, 51, 7084-7099 (* **corresponding authors**)
- Manohar K, Gupta RK, Gupta P, Saha D, S Gare, Sarkar R, **Misra A**, Giri L. FDA approved L-type channel blocker Nifedipine reduces cell death in hypoxic A549 cells through modulation of mitochondrial calcium and superoxide generation. *Free Radical Biology & Medicine*. 2021, 177, 189-200.
- Muleya A, Karumbana KS, Gupta P, Kumbhakara S, Giri B, Raut R, **Misra A***, Maji S*. Synthesis, Structure, Spectro-electrochemical Aspects and Anticancer Activity of Ruthenium (II) Arene Complexes with Substituted Triazole Ligands. *Journal of Organometallic Chemistry*. 2021, Dec 1; 954–955, 122074 (* **corresponding authors**)
- Raut R, Gupta P, Saini T, Mishra P, **Misra A**. Src kinase: An attractive therapeutic target for prostate cancer treatment. *Protein kinase Inhibitors, Elsevier*, 2022, 479-503
- Giri B, Saini T, Kumbhakar S, Kalai Selvan K, Muleya A, **Misra A***, Maji S*. Near-IR Light-induced Photo release of nitric oxide (NO) on Ruthenium Nitrosyl Complexes: Electronic structure, reactivity aspects, and biological effects. *Dalton Transactions* 2020, Aug 11;49(31):10772-10785. (* **corresponding authors**)
- **Misra A*** and Green MR*. Fluorescence reporter-based genome-wide RNA interference screening to identify alternative splicing regulators. *Methods in Molecular Biology* 2017, 1507:1-12 (* **corresponding authors**)
- **Misra A***, Green MR*. From polyadenylation to splicing: Dual role for mRNA 3' end formation factors. *RNA Biology* 2016, 13(3):259-64 (* **corresponding authors**)

- **Misra A***, Jianhong O, Zhu J, Green MR*. Global analysis of CPSF2-mediated alternative splicing: Integration of global iCLIP and transcriptome profiling data. *Genomics Data*. 2015 Dec 6; 217-221 (*** corresponding authors**)
- **Misra A**, Jianhong O, Zhu J, Green MR. Global promotion of alternative internal exon usage by mRNA 3' end formation factors. *Molecular Cell*. 2015 Jun 4;58(5):819-31
- Patra D, Srikalaivani R, **Misra A**, Singh DD, Selvaraj M, Vijayan M. Cloning, expression, purification, crystallization and preliminary X-ray studies of a secreted lectin (Rv1419) from *Mycobacterium tuberculosis*. *Acta Crystallogr Sect F Struct Biol Cryst Commun*. 2010 Dec 1;66(Pt 12):1662-5
- Upadhyay SK, **Misra A**, Surolia N, Surolia A, Sundd M. Backbone chemical shift assignments of the acyl-acyl carrier protein intermediates of the fatty acid biosynthesis pathway of *Plasmodium falciparum*. *Biomol NMR Assign*. 2010 Apr;4(1):83-5. Epub 2010 Feb 25. Erratum in: *Biomol NMR Assign*. 2010 Apr;4(1):87.
- Upadhyay SK*, **Misra A***, Srivastava R, Surolia N, Surolia A, Sundd M. Structural Insights into the acyl-intermediates of *Plasmodium falciparum* fatty acid synthesis pathway; the mechanism of expansion of acyl carrier protein core. *J. Biol. Chem* (2009) Aug 14;284 (33):22390-400. (*** equal contribution**)
- **Misra A**, Surolia N, Surolia A. Evolution of self-acylation property in acyl carrier proteins. *IUBMB Life* (2009) Aug;61(8):853-9.
- **Misra A**, Surolia N, Surolia A. Catalysis and mechanism of malonyl transferase activity in type II fatty acid biosynthesis acyl carrier proteins. *Molecular Biosystems* (2009), 5, 651 – 659
- **Misra A**, Sharma SK, Surolia N, Surolia A. Self-acylation properties of type II fatty acid biosynthesis acyl carrier proteins. *Chemistry & Biology* (2007), 14, 775-83.
- Midha S, Mishra R, Aziz MA, Sharma M, **Mishra A**, Khandelwal P, Bhatnagar R. Cloning, expression, and characterization of recombinant nitric oxide synthase-like protein from *Bacillus anthracis*. *Biochem Biophys Res Commun*. (2005), 336, 346-56.

FUNDING

- SERB-ECR. 49.50 Lakhs, 2018-21, Role-PI
- IITH Interdisciplinary grants. 10.00 Lakhs, 2020-22, Role-Co-PI
- ICMR Gol. ~52.00 Lakhs, 2023-26, Role-PI
- ANRF-CRG. ~67.00 Lakhs, 2023-26, Role-PI
- JICA Fellowship for New Partnership. 5.00 Lakhs, 2023-25, Role-PI
- DBT Gol. ~54.00 Lakhs, 2024-26, Role- Co-PI
- DST-SATHI. ~60.00 Crores, 2024. Role- Co-PI

SELECTED PROFESSIONAL ACTIVITIES

- Editorial Board Member- Discover Oncology (2025 -present)
- Review Editor- Frontiers in Genetics (2021-present)
- Review Editor- Frontiers in Molecular Biosciences (2021-present)
- Ad-hoc Reviewer: Scientific Reports (2017-present)
- Ad-hoc Reviewer: PLOS ONE (2017-present)
- Ad-hoc Reviewer: IUBMB Life (2018-present)

- Ad-hoc Reviewer: Epigenetics & Chromatin (2025-present)
- Ad-hoc Reviewer: International Journal of Experimental Pathology (2025-present)
- Ad-hoc Reviewer: Journal of Physiology and Biochemistry (2025-present)

STUDENT SUPERVISION

- Under Supervision: 7 Ph.D. students (2 PMRF Fellows, 1 DBT-JRF, 1 Interdisciplinary PhD student); 1 MTech student, 1 Project Research Scientist.
- Graduated: 2 PhD Thesis, 9 MTech Thesis

COLLABORATIONS ESTABLISHED

Outside IITH

1. Dr. Manish Bais, Assistant Professor, Department of Translational Dental Medicine, Boston University, Boston, MA, USA (Multiple coauthor publications) (**Research Collaborator**)
2. Prof. Tetsuro Hirose, Professor, Graduate School of Frontier Biosciences, Osaka University. (Received JICA Fellowship for New Partnership for joint research and student visit) (**Research Collaborator**)
3. Dr. Vishal B Rao, Basavatarakam Indo American Cancer Hospital and Research Institute Banjara Hills, Hyderabad (**Clinical Collaborator**) (Coauthor publication)
4. Prof. Rajesh Kashyap, Department of Hematology, Sanjay Gandhi Postgraduate Institute of Medical Sciences, Lucknow (**Clinical Collaborator**)
5. Prof. Nishant Verma, Department of Pediatrics, King George Medical University, Lucknow (**Clinical Collaborator**) (Multiple publications)
6. Prof. Ashutosh Kumar, Professor, Department of Biosciences and Bioengineering, IIT Bombay (**Research Collaborator**)

Inside IITH with faculty from other departments

1. Dr. Somnath Maji, Dept. of Chemistry, IITH (Multiple collaborative publications)
 2. Dr. Suhanya Duraiswamy, Department of Chemical Engineering, IITH (Co-PI on research grants)
 3. Dr. Lopamudra Giri, Department of Chemical Engineering, IITH (Collaborative publication)
 4. Dr. Satyavrata Samavedi, Department of Chemical Engineering, IITH (Co-supervising ID PhD student, PMRF fellow)
 5. Dr. Shishir Kumar, Department of Electrical Engineering, IITH (ID grant)
-